

## PORTAGE LA PRAIRIE SOUTHPORT, MB, Canada

WMO: 718510

Lat: 49.9031N

Lon: 98.2839W

Elev: 273

StdP: 98.09

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |              |             |              | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|--------------|-------------|--------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |              | 1%          |              |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB         | WS          | MCDB         | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-30.0 | (c)<br>-27.5 | (d)<br>-34.1                  | (e)<br>0.2 | (f)<br>-29.7 | (g)<br>-31.6 | (h)<br>0.2 | (i)<br>-27.2 | (j)<br>13.2           | (k)<br>-14.6 | (l)<br>11.8 | (m)<br>-14.7 | (n)<br>3.0            | (o)<br>240 | (p)<br>0.642 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>10.6            | (c)<br>30.1     | (d)<br>21.5 | (e)<br>28.3 | (f)<br>20.5 | (g)<br>26.7 | (h)<br>19.3 | (i)<br>23.4         | (j)<br>27.8 | (k)<br>21.9 | (l)<br>26.4 | (m)<br>20.7 | (n)<br>25.1 | (o)<br>4.4           | (p)<br>180 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 22.0                            | 17.2 | 26.3 | 20.4 | 15.6 | 24.6 | 19.1 | 14.4 | 23.2 | 71.3          | 28.0 | 65.7 | 26.3 | 61.2 | 25.1 | 28.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 11.3              | 9.8  | 8.6 | -33.7                      | 33.6 | 3.0                | 1.8 | -35.8                                              | 34.9 | -37.5      | 35.9 | -39.2      | 36.9 | -41.4      | 38.2 |
|                   |      |     | -33.7                      | 26.2 | 3.0                | 1.5 | -35.8                                              | 27.3 | -37.6      | 28.1 | -39.3      | 29.0 | -41.5      | 30.0 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 3.7         | -14.8 | -12.7 | -5.3  | 4.0   | 11.2  | 17.2  | 20.0  | 18.9  | 13.7  | 5.9   | -3.5  | -11.6 | (6)  | (7)  |
|                                                                        | DBStd       | 13.65       | 8.16  | 7.55  | 7.67  | 6.18  | 4.96  | 3.64  | 3.10  | 3.29  | 4.40  | 4.86  | 6.65  | 7.67  | (7)  | (8)  |
|                                                                        | HDD10.0     | 3356        | 770   | 637   | 474   | 194   | 46    | 1     | 0     | 0     | 15    | 146   | 404   | 669   | (8)  | (9)  |
|                                                                        | HDD18.3     | 5519        | 1028  | 870   | 732   | 430   | 228   | 62    | 19    | 34    | 149   | 386   | 654   | 927   | (9)  | (10) |
|                                                                        | CDD10.0     | 1041        | 0     | 0     | 1     | 14    | 82    | 217   | 309   | 274   | 126   | 18    | 1     | 0     | (10) | (11) |
|                                                                        | CDD18.3     | 163         | 0     | 0     | 0     | 0     | 5     | 28    | 69    | 50    | 9     | 1     | 0     | 0     | (11) | (12) |
|                                                                        | CDH23.3     | 1468        | 0     | 0     | 0     | 3     | 81    | 259   | 558   | 453   | 107   | 8     | 0     | 0     | (12) | (13) |
|                                                                        | CDH26.7     | 364         | 0     | 0     | 0     | 0     | 18    | 58    | 140   | 126   | 22    | 1     | 0     | 0     | (13) | (14) |
| (14) Wind                                                              | WSAvg       | 3.9         | 4.1   | 4.1   | 4.2   | 4.5   | 4.4   | 3.6   | 3.3   | 3.3   | 3.7   | 4.1   | 4.1   | 3.9   | (14) | (15) |
| (15) Precipitation                                                     | PrecAvg     | 510         | 19    | 16    | 29    | 34    | 69    | 85    | 70    | 57    | 43    | 40    | 24    | 24    | (15) | (16) |
|                                                                        | PrecMax     | 727         | 45    | 56    | 71    | 81    | 157   | 143   | 164   | 125   | 98    | 106   | 99    | 47    | (16) | (17) |
|                                                                        | PrecMin     | 343         | 7     | 0     | 5     | 9     | 18    | 40    | 15    | 13    | 5     | 0     | 0     | 2     | (17) | (18) |
|                                                                        | PrecStd     | 86          | 9     | 14    | 19    | 18    | 33    | 31    | 40    | 27    | 22    | 26    | 21    | 13    | (18) | (19) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 5.0   | 5.6   | 15.2  | 23.4  | 29.2  | 31.2  | 32.1  | 32.3  | 29.2  | 24.0  | 15.0  | 5.9   | (19) | (20) |
|                                                                        |             | MCWB        | 2.7   | 2.5   | 9.4   | 12.9  | 17.7  | 22.1  | 23.5  | 22.5  | 19.9  | 15.2  | 8.9   | 2.8   | (20) | (21) |
|                                                                        | 2%          | DB          | 2.5   | 2.8   | 10.2  | 19.8  | 25.6  | 28.2  | 29.9  | 29.8  | 26.5  | 19.2  | 10.7  | 3.4   | (21) | (22) |
|                                                                        |             | MCWB        | 0.8   | 1.2   | 6.0   | 11.1  | 14.7  | 19.4  | 22.2  | 21.3  | 18.3  | 12.3  | 6.5   | 1.3   | (22) | (23) |
|                                                                        | 5%          | DB          | 0.0   | 0.8   | 6.8   | 17.1  | 22.7  | 26.2  | 28.1  | 27.8  | 23.9  | 16.2  | 7.7   | 0.6   | (23) | (24) |
|                                                                        |             | MCWB        | -1.1  | -0.2  | 4.2   | 9.9   | 13.8  | 18.6  | 21.3  | 20.3  | 17.3  | 11.0  | 4.5   | -0.8  | (24) | (25) |
|                                                                        | 10%         | DB          | -2.8  | -1.3  | 4.2   | 14.1  | 20.0  | 24.3  | 26.5  | 25.9  | 21.3  | 13.5  | 5.1   | -1.3  | (25) | (26) |
|                                                                        |             | MCWB        | -3.7  | -2.3  | 2.4   | 8.2   | 12.4  | 17.7  | 20.1  | 19.3  | 15.8  | 9.7   | 3.1   | -2.2  | (26) | (27) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 2.8   | 3.0   | 10.0  | 14.3  | 18.7  | 24.5  | 25.9  | 25.0  | 22.1  | 16.1  | 9.9   | 3.1   | (27) | (28) |
|                                                                        |             | MCDB        | 4.8   | 5.0   | 14.9  | 20.5  | 25.9  | 29.3  | 29.8  | 29.5  | 26.2  | 22.7  | 14.0  | 5.5   | (28) | (29) |
|                                                                        | 2%          | WB          | 1.0   | 1.3   | 6.5   | 11.8  | 16.6  | 21.6  | 23.8  | 22.7  | 19.8  | 13.6  | 6.8   | 1.4   | (29) | (30) |
|                                                                        |             | MCDB        | 2.3   | 2.6   | 10.0  | 18.4  | 21.6  | 25.6  | 27.9  | 27.8  | 24.4  | 17.9  | 9.8   | 3.3   | (30) | (31) |
|                                                                        | 5%          | WB          | -1.1  | 0.0   | 4.2   | 10.2  | 15.0  | 20.0  | 22.4  | 21.3  | 18.2  | 11.7  | 4.8   | -0.5  | (31) | (32) |
|                                                                        |             | MCDB        | 0.0   | 1.0   | 6.7   | 15.9  | 20.3  | 24.2  | 26.5  | 26.2  | 22.3  | 15.5  | 7.4   | 0.6   | (32) | (33) |
|                                                                        | 10%         | WB          | -3.7  | -2.3  | 2.5   | 8.7   | 13.6  | 18.6  | 21.1  | 20.1  | 16.7  | 9.8   | 3.1   | -2.1  | (33) | (34) |
|                                                                        |             | MCDB        | -2.9  | -1.3  | 4.3   | 14.1  | 18.8  | 22.8  | 25.4  | 24.5  | 20.6  | 13.4  | 5.1   | -1.2  | (34) | (35) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 8.5   | 8.9   | 9.0   | 11.0  | 11.5  | 10.5  | 10.6  | 11.6  | 11.0  | 9.0   | 7.8   | 7.7   | (35) | (36) |
|                                                                        |             | MCDBR       | 10.2  | 9.5   | 11.3  | 16.1  | 15.8  | 13.5  | 12.6  | 14.5  | 15.2  | 14.2  | 11.8  | 9.5   | (36) | (37) |
|                                                                        | 5% WB       | MCWBR       | 9.1   | 8.1   | 8.0   | 9.3   | 8.1   | 7.2   | 6.9   | 7.7   | 8.7   | 8.6   | 8.1   | 8.0   | (37) | (38) |
|                                                                        |             | MCWBR       | 10.0  | 9.3   | 10.7  | 14.9  | 12.6  | 11.7  | 11.9  | 12.9  | 13.4  | 13.4  | 11.0  | 9.2   | (38) | (39) |
| (39) Clear-Sky Solar Irradiance                                        | taub        |             | 0.262 | 0.282 | 0.299 | 0.356 | 0.374 | 0.378 | 0.388 | 0.381 | 0.347 | 0.300 | 0.268 | 0.248 | (39) | (40) |
|                                                                        |             | taud        | 2.267 | 2.228 | 2.342 | 2.334 | 2.345 | 2.364 | 2.349 | 2.358 | 2.445 | 2.549 | 2.467 | 2.347 | (40) | (41) |
|                                                                        | Ebn at Noon |             | 785   | 858   | 912   | 888   | 884   | 880   | 864   | 852   | 850   | 845   | 792   | 761   | (41) | (42) |
|                                                                        |             | Edh at Noon | 69    | 93    | 101   | 115   | 118   | 117   | 117   | 111   | 92    | 69    | 57    | 55    | (42) | (43) |
| (43) All-Sky Solar Radiation                                           | RadAvg      |             | 1.41  | 2.60  | 3.91  | 5.03  | 5.53  | 5.91  | 6.25  | 5.16  | 3.73  | 2.24  | 1.24  | 1.01  | (43) | (44) |
|                                                                        | RadStd      |             | 0.12  | 0.19  | 0.31  | 0.35  | 0.46  | 0.41  | 0.26  | 0.36  | 0.28  | 0.29  | 0.17  | 0.11  | (44) | (45) |

## Historical Trends

|      |                         | Heating |        | Cooling |       |       | Degree-Days |         |         |         |     |      |
|------|-------------------------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|-----|------|
|      |                         | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |     |      |
|      |                         | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     | (m) |      |
| (46) | Station Only            | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | +70     | N/A | (46) |
| (47) | Regional (25 neighbors) | N/A     | N/A    | N/A     | N/A   | -0.56 | -0.78       | N/A     | N/A     | N/A     | N/A | (47) |

Nomenclature: See separate page